

Table 6: GSM8K Question Answering GPT4-Judge

Cache Budget / Fastgen Attn Recovery Frac (%)	Strategy	Similarity to GT	Coherence	Faithfulness	Helpfulness
10	HASHEVICT	1.018	1.387	1.147	1.083
	L_2	1.005	1.293	1.098	1.033
	H ₂ O	1.172	2.304	1.566	1.630
	Scissorhands	1.138	2.132	1.424	1.452
30	HASHEVICT	2.520	3.767	3.216	3.190
	L_2	1.356	2.428	1.895	1.841
	H ₂ O	3.014	4.252	3.706	3.860
	Scissorhands	2.906	4.184	3.636	3.798
50	HASHEVICT	3.457	4.530	4.212	4.241
	L_2	2.190	3.494	3.035	3.027
	H ₂ O	3.798	4.712	4.434	4.534
	Scissorhands	3.582	4.604	4.276	4.400
70	HASHEVICT	3.734	4.671	4.404	4.444
	L_2	2.934	4.184	3.817	3.820
	H ₂ O	3.940	4.774	4.576	4.656
	Scissorhands	3.712	4.668	4.334	4.462
90	HASHEVICT	3.569	4.578	4.324	4.361
	L_2	3.837	4.722	4.468	4.525
	H ₂ O	3.970	4.814	4.596	4.688
	Scissorhands	3.750	4.676	4.392	4.504
50	Fastgen	1.000	1.074	1.040	1.028
60		1.000	1.054	1.022	1.010
70		1.008	1.116	1.048	1.014
80		1.472	2.602	2.118	2.234
90		3.838	4.714	4.448	4.554
100	Full	3.845	4.716	4.499	4.545

Compressing the KV cache entails dropping KV pairs. Per [Devoto et al., 2024], we can define the attention loss caused by the compression as the sum of the attention scores associated with the dropped KV pairs in layer l and head h via the equation $L_{l,h}^m = \sum_{p \in D_{l,h}^m} a_{l,h,p}$, where $a_{l,h,p}$ is the average attention score at position p for layer l and head h , and $D_{l,h}^m$ denotes the positions of the m dropped KV pairs, with $|D_{l,h}^m| = m$. We process a selection of prompts and examine how proposed evictions by the L_2 eviction strategy and HASHEVICT would affect the sum of attention scores.

To quantify the additional attention loss introduced by using an alternative ranking method (such as L_2 norm or HASHEVICT’s F_{score}) instead of the true attention-based ranking, we define the cumulative attention loss difference as:

$$Y_{l,h} = \sum_{m=1}^n (L_{l,h}^m - L_{l,h,\text{ref}}^m), \quad (7)$$

where $L_{l,h,\text{ref}}^m$ is the cumulative attention loss when dropping the KV pairs with the actual lowest attention scores. The value $Y_{l,h}$ is non-negative, and a lower value indicates that the ranking method closely approximates non-compressed attention. Figure 8 depicts the ALR for the L_2 eviction rankings and an LSH ranking.

It is important to note that HASHEVICT is not designed to produce a global ranking among the keys as the L_2 method is designed to do (via a low-to-high ordering of all L_2 key norms). HASHEVICT ranks the importance of past tokens with regards to the current token – and this ranking changes every step. To simulate a comparison, we record the average Hamming distance between the key code of token i and the query codes of all tokens $j > i$. We then sort tokens from lowest to highest average Hamming distance. Figure 8a reflects the ALR according to this ranking system. The L_2 ranking exclusively prefers high-attention tokens, while the LSH ranking prefers medium-to-high-attention tokens. Based on our empirical results in Section 4, the selection of tokens over a spectrum of attention scores skewing towards high results in greater task versatility compared to the L_2 eviction.

Table 7: MedQA Question Answering BERTScore

Cache Budget / Fastgen Attn Recovery Fra(%)	Strategy	Precision	Recall	F1
10	HASHEVICT	0.857	0.808	0.832
	L_2	0.833	0.813	0.823
	H ₂ O	0.866	0.795	0.829
	Scissorhands	0.867	0.795	0.829
30	HASHEVICT	0.867	0.834	0.850
	L_2	0.855	0.834	0.844
	H ₂ O	0.878	0.802	0.838
	Scissorhands	0.877	0.802	0.838
50	HASHEVICT	0.869	0.842	0.855
	L_2	0.866	0.841	0.853
	H ₂ O	0.879	0.803	0.839
	Scissorhands	0.878	0.804	0.839
70	HASHEVICT	0.869	0.843	0.855
	L_2	0.868	0.842	0.855
	H ₂ O	0.879	0.801	0.838
	Scissorhands	0.879	0.803	0.839
90	HASHEVICT	0.868	0.843	0.855
	L_2	0.868	0.843	0.855
	H ₂ O	0.879	0.801	0.838
	Scissorhands	0.880	0.802	0.839
50	Fastgen	0.816	0.763	0.788
60		0.806	0.766	0.785
70		0.811	0.766	0.787
80		0.866	0.793	0.828
90		0.876	0.800	0.836
100	Full	0.868	0.843	0.855

E On the Analysis of the Relationship between Attention Scores and Average, Accumulated LSH Hamming Distance

In this section, we follow up on our ALR in Appendix Section D. We analyze the relationship between attention scores and average LSH Hamming distances using 50 randomly selected prompts from GSM8K. We stress that this metric does not perfectly capture the "ranking" system of HASHEVICT (which cannot perform a global/full-sequence token-importance ranking like L_2 eviction).

For each prompt, we performed the following:

1. **Captured States:** Extracted normalized key and query vectors from every layer and head combination after applying rotary positional embeddings.
2. **Applied Random Projections:** Applied multiple random Gaussian projections, varying the projection length (number of bits). We tested with projection lengths of 8, 16, 24, and 32.
3. **Computed Hamming Distances:** Computed the Hamming distances between the projected and binarized vectors and averaged this over multiple projections to mitigate the randomness that LSH introduces and to obtain a more stable estimate of the Hamming distances.
4. **Computed Correlations:** Calculated the Pearson correlation coefficient between the attention scores and the inverted average Hamming distance for each layer and head combination and for each projection length.

Table 8: MedQA Question Answering Rouge

Cache Budget (%)	Strategy	Rouge 1	Rouge 2	Rouge L	Rouge Lsum
10	HASHEVICT	0.346	0.110	0.171	0.324
	L_2	0.304	0.072	0.154	0.289
	H ₂ O	0.236	0.092	0.138	0.220
	Scissorhands	0.237	0.091	0.139	0.221
30	HASHEVICT	0.449	0.170	0.227	0.426
	L_2	0.429	0.146	0.213	0.407
	H ₂ O	0.255	0.118	0.151	0.239
	Scissorhands	0.252	0.116	0.151	0.236
50	HASHEVICT	0.481	0.194	0.245	0.455
	L_2	0.474	0.184	0.240	0.449
	H ₂ O	0.243	0.107	0.149	0.229
	Scissorhands	0.244	0.110	0.150	0.230
70	HASHEVICT	0.487	0.197	0.249	0.461
	L_2	0.484	0.194	0.247	0.458
	H ₂ O	0.229	0.097	0.143	0.216
	Scissorhands	0.234	0.103	0.147	0.219
90	HASHEVICT	0.487	0.197	0.249	0.461
	L_2	0.487	0.197	0.249	0.461
	H ₂ O	0.223	0.095	0.142	0.211
	Scissorhands	0.228	0.099	0.145	0.214
50	Fastgen	0.068	0.013	0.052	0.066
60		0.079	0.014	0.061	0.077
70		0.103	0.020	0.074	0.099
80		0.208	0.069	0.126	0.192
90		0.220	0.092	0.140	0.207
100	Full	0.486	0.198	0.248	0.460

E.1 Results

The average Pearson correlation between the attention scores and the inverted average Hamming distances is 0.2978 ± 0.1947 . Table 11 and Figure 9a detail the average Pearson correlation per projection length.

E.2 Observations

- **Correlation with Projection Length:** As shown in Figure 9a and Table 11 the average Pearson correlation increases with projection length. This is likely due to the more detailed vector representation in the projected space, allowing for finer-grained similarity comparisons.
- **Layer-wise Trends:** Figure 9b shows a slight decrease in the average Pearson correlation for the later transformer layers. Earlier layers may be more focused on recognizing broader patterns where the similarity LSH captures is more pronounced compared to the latter layers, which may focus on specifics not captured as effectively by Hamming distances.
- **Head-wise Consistency:** The correlation between attention scores and inverted average Hamming distance is relatively consistent across different attention heads, with little variance as seen in Figure 9c. This uniform behavior indicates that the relationship between attention scores and LSH-measured similarity is, to a large extent, independent of specific head functions.
- **LSH vs. L_2 Norms:** While L_2 norms were more effective at identifying high-attention tokens, LSH excelled at identifying tokens with moderate attention scores that are vital for the generation of coherent language output. This aligns with the findings of Guo et al. [2024], which suggests that tokens with low to medium attention scores are crucial for high-quality language generation.